

RANDOMIZED MINCUT

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Problem Overview:

The problem is to find the minimum cut of an undirected graph, which is the smallest number of edges whose removal disconnects the graph.

Solution approach:

1. Read the number of vertices and edges from the input file
2. Store all edges in a list
3. While the number of vertices is greater than two, randomly select an edge (using `uniform_int_distribution`) and merge its two vertices into one
4. Update all edges to reflect the merged vertex
5. Remove any self-loops created during the merge
6. Decrease the total vertex count by one
7. When only two vertices remain, count the edges between them
8. The count represents the cut size for that run
9. Write the result to the output file
10. Repeat the process multiple times to increase the chance of finding the minimum cut